4.2.8. Titanic Dataset Analysis and Data Cleaning - 4

26:56

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Get the number of survivors by gender (Sex).
2. Get the number of non-survivors by gender (Sex).
3. Get the number of survivors by embarkation location (Embarked_S).
4. Get the number of non-survivors by embarkation location (Embarked_S).
5. Calculate the percentage of children (Age < 18) who survived.
6. Calculate the percentage of adults (Age >= 18) who survived.
7. Get the median age of survivors.
8. Get the median age of non-survivors.
9. Get the median fare of survivors.
10. Get the median fare of non-survivors.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina
Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

Note: Refer to the visible test case for better reference.

Code:

```
import pandas as pd

import numpy as np
```

```
# Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)


# 1. Get the number of survivors by gender

survivors_by_gender = data[data['Survived'] == 1]['Sex'].value_counts()

print(survivors_by_gender)


# 2. Get the number of non-survivors by gender

non_survivors_by_gender = data[data['Survived'] == 0]['Sex'].value_counts()

print(non_survivors_by_gender)


# 3. Get the number of survivors by embarked location (Embarked_S)

survivors_by_embarked_s = data[data['Survived'] == 1]['Embarked_S'].value_counts()

print(survivors_by_embarked_s)


# 4. Get the number of non-survivors by embarked location (Embarked_S)

non_survivors_by_embarked_s = data[data['Survived'] == 0]['Embarked_S'].value_counts()

print(non_survivors_by_embarked_s)


# 5. Percentage of children (Age < 18) who survived

children = data[data['Age'] < 18]

children_survival_rate = children['Survived'].mean()

print(children_survival_rate)
```

```python
# 6. Percentage of adults (Age >= 18) who survived

adults = data[data['Age'] >= 18]

adults_survival_rate = adults['Survived'].mean()

print(adults_survival_rate)


# 7. Median age of survivors

median_age_survivors = data[data['Survived'] == 1]['Age'].median()

print(median_age_survivors)


# 8. Median age of non-survivors

median_age_non_survivors = data[data['Survived'] == 0]['Age'].median()

print(median_age_non_survivors)


# 9. Median fare of survivors

median_fare_survivors = data[data['Survived'] == 1]['Fare'].median()

print(median_fare_survivors)


# 10. Median fare of non-survivors

median_fare_non_survivors = data[data['Survived'] == 0]['Fare'].median()

print(median_fare_non_survivors)
```

Explorer

Average time	Maximum time
0.417 s	**0.417 s**
417.00 ms	417.00 ms

1 out of 1 shown test case(s) passed

Test case 1 417 ms Debug

Expected output	Actual output
female····233	female····233
male······109	male······109
Name:·Sex,·dtype:·int64	Name:·Sex,·dtype:·int64
male······468	male······468
female·····81	female·····81
Name:·Sex,·dtype:·int64	Name:·Sex,·dtype:·int64
1····217	1····217
0····125	0····125
Name:·Embarked_S,·dtype:·int64	Name:·Embarked_S,·dtype:·int64
1····427	1····427
0····122	0····122
Name:·Embarked_S,·dtype:·int64	Name:·Embarked_S,·dtype:·int64
0.5398230088495575	0.5398230088495575
0.3810316139767055	0.3810316139767055
28.0	28.0
28.0	28.0
26.0	26.0
10.5	10.5